Progress Report 1

1. **Project title:**
   Swarm of robots to complete locomotive and object-searching tasks, and touchscreen interface for a multipurpose robot.

2. **Names, student numbers, and emails**:
   Group 36 members: (all members are registered in ECSE 458-D1)

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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**
   All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

   Weekly meeting with the entire group: Thursdays at 4pm
   Weekly meeting with the advisor: Fridays at 2pm

   Outlined below is a detailed list of the meetings that have been attended with the advisor, including dates and any key discussions or decisions made during each session.
   Group meeting(s)
   - Friday, 20th September
     - The group members engaged in a detailed discussion about the various available projects
     - We analysed the pros and cons of each project, considering different factors such as feasibility, complexity, and impact.
     - Each member expressed their interest in specific aspects or components of the projects.
     - Their interests spanned across different areas of the available projects, reflecting a diverse set of preferences and strengths
     - Selected between two project options.

   Meeting(s) with Advisor
   - Tuesday, 1st October

- The team members had an elaborate discussion into the project's scope, objectives, and technical requirements. We explored the key milestones, deliverables, and the division of tasks among each member.
- We finalised the meeting schedule, choosing platforms for virtual meetings (e.g., Zoom, Teams) and confirming the lab space for in-person sessions.
- The advisor informed us of the lab location for project experiments, and we discussed access to equipment and scheduling for shared resources.

Recent progress

During the first few weeks of the project, the team focused on thoroughly exploring and understanding the scope of the project, ensuring we had a clear grasp of the expectations set for us. At the same time, we worked on establishing the necessary logistical arrangements to efficiently organise and execute the tasks ahead. We first started by meeting with our advisor, to set the goals of the project for each semester and to get the contact information of the other two groups whose projects we will be working on.

As such we categorised the projects into two parts spread over each semester:

A. Touchscreen Interface for R1-D4 (Fall 2024): Begin by enhancing R1D4, creating a touchscreen or tablet-based user friendly interface that will allow users to interact with the robot through visual cues, such as facial expressions indicating the robot's status. This interface will not only serve as an emotional feedback system but also allow for manual control over the robot's actions, potentially migrating some server controls to the touchscreen.
B. Swarm Robots (Winter 2025): Optimising the communication between the swarm robots via a central Wi-Fi server and improving their ability to collaborate on tasks. Each robot will receive hardware upgrades, including the addition of sensors and possibly cameras, to autonomously detect and transport objects in an open space. We plan to integrate AI-based algorithms to manage both individual robot actions and collective behaviour within the swarm.

Future plans

- Schedule times to meet with other capstone groups - to be completed by 25 October
- Submit two pdfs which outline the short-term goals and objectives for each semester - to be completed by 25 October
- Understand the other two groups progress so far - to be completed by 1 November
- Identify the software and hardware we will use for the touchscreen interface - to be completed by 8 November
- Come up with a plan to implement the user interface - to be completed by 8 November
- Assign tasks and roles to each team member - to be completed by 8 November

Lifelong Learning

Here is a list of some skills that we will need to acquire to complete this project:

- Better time management within our group. We need to be able to plan regular meetings and stick to a constituent schedule so we don't fall behind.
- Become more familiar with the current code base of the R1-D4 robot and its behaviour so we can begin to classify its actions.
- Figure out what type of GUI libraries are available that we can use to create the LED interface.
- Learn more about the end goal for the R1-D4 robot so we can plan the control flow of the LED screen accordingly.

Here is what we have learned about self learning so far:

- Make sure we set lots of small intermediate goals so we can easily hit them and we give ourselves a more concrete sense of accomplishment as we pursue the project. This awareness will help prevent us from procrastinating.
- Constant communication between team members is essential so we all know where we are in the project.
- Self learning only works if you have self discipline, so practising that is going to be important as time moves on.